Review of "DualHash: A Stochastic Primal-Dual Algorithm with Theoretical Guarantee for Deep Hashing"

The aim of this article is to propose a new primal-dual stochastic approach, called DualHash, to address the challenge of 'deep hashing' encoding within a non-convex optimisation framework. This has led to the development of two algorithms, DualHash-StoM and DualHash-Stom, based respectively on the principle of momentum acceleration and on a variance reduction technique. The authors present a complexity analysis of the two schemes before conducting experiments to compare their performances with existing methods on image datasets. The proposed article is relatively clear and well-referenced, with a comprehensive introduction. Its strongest and most relevant being the one dedicated to numerical experiments well detailled. However, the theoretical sections contain several weaknesses that need to be addressed in order to fully meet the journal's requirements. For all these reasons, **I return the article to the authors for major revisions.**

General comments

- In some places, you use of letter d_x to denote the space dimension relative to variable x and sometimes letter q instead. I would prefer the use of letter q everywhere, d_x notation suggests that the space dimension may change depending on the value of x which is a bit confusing.
- Some important equations/formulations need to be referenced (see the detailed remarks).
- h and h^* functions are used for vectors and matrix objects. This is an acceptable abuse of notation in the context of your work provided that this is clearly stated, for example in your notation subsection.
- Equalities or inequalities involving random variables (typically those containing conditional expectations) are 'only' verified almost surely; this is why it is important to specify this whenever necessary.
- Some 'non-matrix' variables (for example, $\boldsymbol{D}^k$, $\boldsymbol{G}^k$) are written in capital letters, which does not strictly adhere to the conventions you set out in the notations subsection.
- Some proofs are placed in the main paper and others similar in the appendix (for example Lemma 5.4/Lemma 5.5). Is there a specific reason for this ?

1 Abstract/Introduction

Abstract and introduction appear generally well-written and well-cited to me . It is clear what the authors are trying to achieve in this paper.

- (page 1) "W-type regularization such as $z \mapsto ||z| - 1|$".
- (page 1) I would reformulate as "Mathematically, this amounts **to carry out** a projection onto [...]. **Such a process can be interpreted as** a nonconvex discrete optimization problem[...], where the date fidelity criterion **is based on** pairwise similarity..."
- (page 1) (Typo ?) "up to 512×"

- (page 1) "such as $z \mapsto ||z| - 1|$"
- (page 2) "using subgradients may result in slow convergence and oscillation." add references to illustrate such a fact.
- (page 2) "inverse imaging problems" I would also add additional references, in the vein of [3] et al.
- (page 2) "its composition with neural network outputs $h(\mathbf{D}i(x))$" sounds a bit unclear to me.
- (page 2) "when h is component-wise nondecreasing", to verify but after quickly checked [29], I think h also need to be convex. This would corroborate your next sentence "when h itself is nonconvex...".
- (page 2-3) "reformulate the problem as **one of constrainted optimization**".
- (page 3) "[...] deep hashing; directly extending their results to our setting is nontrivial." I would cut the sentence in two, for example : "[...] deep-hashing. A direct extension of their results to our setting is far from being trivial. "
- (page 3) "the challenging composite term $h \circ \mathbf{D}$"
- (page 4) Regarding to Table 1.1, your paper seems to be the first one to conduct a complexity analysis for deep hashing methods. Are you totally sure of this ?
- (page 4/Related work) "W-type regularizer such as $z \mapsto \||z|-1\|_1$", "employ $z \mapsto \||z|-1\|_2^2$ as a smooth surrogate for $z \mapsto \||z| - 1\|_1$".
- (page 5) "and ~~the~~ stochastic majorization-minimization subspace methods".
- (page 5) I would give a number to the equation relative the reformulation of Problem (1.1). Also explain what the functions f and c are supposed to represent (for example is $f = 1/n \sum f_i$? ...).

2 Preliminaries

- (page 5) Your definition of closeness for f is equivalent to those of the lower semicontinuity (see for example chapter 1 of [1]. In the appendix, you mention the two terms (closeness in Def A.1, lower semicontinuity Prop A.2) . Because the second one, appears more popular to me, I would suggest to work with lower semicontinuity everywhere.
- (page 6) Wouldn't it be "$f(\boldsymbol{x}) = \mathbb{E}_\epsilon[F(\boldsymbol{x}, \epsilon)]$" instead ? Moreover, what function $F : (x, \epsilon) \mapsto F(x, \epsilon)$ is supposed to represent, is this a generic one ?
- (page 6) Remind the reader that $\mathbf{d}^k$ correspond to the current direction at iteration k (you might even recall the generic scheme $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \eta_k \boldsymbol{d}^k$). Moreover, give a number to the equation relative to $\boldsymbol{d}^k$ update's rule for StoRM algorithm.
- (page 6) "A differentiable function $f : \mathbb{R}^q \to \mathbb{R}$ is said to be L_f-smooth".
- (page 6) Add reference [2] for descent lemma.

3 Problem reformulation

- (page 6) Give a number to the equation relative to deep hashing + typo under the min " $x \underline{\in} \mathbb{R}^q$ ".
- (page 7) Variable $\boldsymbol{b}_i$ denotes first $sgn(\boldsymbol{D}_i(\boldsymbol{x}))$ and then, in subsection 3.2, corresponds to $\boldsymbol{D}_i(\boldsymbol{x})$, please clarify with two separated notations.
- (page 7) If I have clearly understood, problem (P) is a regularized version of (3.1) which is supposed to better leverage the structure of the initial problem. However, there seems to be a confusion relative to some variables definitions otherwise I don't really understand the point of introducing $P(\boldsymbol{x}, \boldsymbol{B})$, (since $\boldsymbol{b}_i = \boldsymbol{D}_i(\boldsymbol{x})$, we have $\|\boldsymbol{b}_i - \boldsymbol{D}_i(\boldsymbol{x})\|^2 = 0$ and therefore $P(\boldsymbol{x}, \boldsymbol{B}) = 0$). Finally, could you provide further details which would explain the interest of term $P(\boldsymbol{x}, \boldsymbol{B})$ (it looks like a coupling one at first glance), don't hesitate to use references if necessary.

4 Stochastic Primal-Dual Hashing Algorithms

- (page 7) I would place definition in (4.1) in the subsection notations. Do not forget to mention the fact you use the same notations for both vectors and matrices (see 4th remark of "General comments" section)
- (page 8) What do you mean by "inexpensive gradient" ?
- (page 9) In (4.5) first line, to avoid any confusion I would write "$(\tau^{-1}h^*)$" instead of "$(1/\tau h^*)$". Moreover, I think a $1/\tau$ factor is missing before $(2\boldsymbol{B}^{k+1} - \boldsymbol{B}^k)$.
- (Algorithms 4.1,4.2) Reading through your pseudo-code, one might think the choice of every batch size is determined before the algorithm is run. In the case of a constant batch size $b_k = b$, that makes perfect sense otherwise, it seems a bit strange to me.

5 Complexity analysis

This section brings together a fairly large number of results, including intermediate lemmas and theorems, which may seem a bit exhaustive and even disjoined as it forces the reader to keep referring back and forth to the appendices. To such an extent, my comments relative to proofs postponed in Appendix C or D are covered in a dedicated section, at the end of this document. A few mathematical inaccuracies needs to be reworked and particularly the stochastic formalism of the paper needs to be a little refined. Finally, your complexity analysis involves a lot of parameters whose values tend to vary depending on the subsections, which can sometimes detract from the main point.

- (page 10) "the set crit($\mathscr{L}$)... is nonempty", I would place this fact in Assumption 5.1.
- (Assumption 5.1) I would reformulate (ii) as " F is L_F-smooth ($L_F \in (0, +\infty)$) over $\mathbb{R}^q \times \mathbb{R}^{n \times d}$; ".
- (Assumption 5.1) In (iii), you use several notations that you forgot to introduce earlier, in particular the conditional expectation operator $\mathbb{E}_i$. To do so, you also need to clearly define a probability space and a filtration. That is why I would tend to add an extra paragraph and/or move here the one you wrote at the very beginning of the Appendix C.

- (Remark 5.1) "The smoothness constant L_F remains $\mathcal{O}(1)$", useless comment, a non-zero constant being always a $\mathcal{O}(1)$.

- (Lemma 5.1/proof) Having J coercive and lower-bounded is generally not enough to directly obtain the boundedness of $(\boldsymbol{x}^k, \boldsymbol{B}^k)$. You classically also need to prove that sequence $\{J(\boldsymbol{x}^k, \boldsymbol{B}^k)\}$ converges (or at least its boundedness). Note also that having boundedness of all level-sets is equivalent to coercivity. Finally, the coercivity of J is not assumed in your assumption 5.1.

- (Lemma 5.2/proof) Some objects as G function is not explicitly defined, making your proof a bit hard to follow. Moreover, which subdifferential property do you use to conclude ? To rework.

- (page 11) "The presence [...] renders the Lagrangian (4.2) non-descent", I understand what you're saying, but the wording is still a bit vague. I would reformulate as "The presence of dual variables Λ makes it difficult to establish a descent property for the Lagrangian".

- (page 12) In the definition of Δ_B^k, there is no need to specify the 'F' for the norm as there is no ambiguity here.

- (Lemma 5.4/proof) I don't really understand how you reach inequality (5.9). Provide far more details (intermediate inequalities, mention descent lemma whenever it is used, how you deal with the conjugate quantities...)). Assuming (5.9) holds, the rest of the proof is correct. Moreover, why did you choose to include this proof here while those of its counterpart (iteration progress on $\boldsymbol{x}$) is presented in the appendix ?

- (page 13) Give more details to justify $\mathbb{E}[\|\boldsymbol{e}^k\|^2] \leq \sigma^2/b_k$.

- (Lemma 5.5/proof) See my comments relative to Appendix C.

- (page 13) "To handle the oscillation of (4.2). I would reformulate as "To handle the oscillations of the Lagrangian function $\mathscr{L}$".

- (page 13) For better consistency and since you state that C_1, C_2, C_3, C_4 are independent of k, you should introduce (5.14) setup before (5.12) and (5.13).

- (page 13) "descent property of $\underline{\{}\Psi_m(\mathcal{Q}^k)\underline{\}}$".

- (page 13) "<u>are</u> established in Lemma C.2".

- (Lemma 5.6/proof) See my comments relative to Appendix C.

- (Lemma 5.7) Why $R \in 2, ..., T+1$ instead of $R \in 1, ..., T$? Furthermore, I am not particularly in favour of using uniform sampling to present a rate; I would prefer to use the quadratic metric $\frac{1}{T}\sum_{k=1}^{T}\mathbb{E}[dist^2(...)]$. Finally, to use Ψ_m^* you need at least to justify the lower-boundedness of $\{\Psi_m(\mathcal{Q}^k)\}$ which doesn't seem straightforward.

  I would first introduce L_Δ before mentioning S_1, S_2, S_3, S_4 (these constant actually don't appear in (5.17) as such).

- (Lemma 5.7/proof) See my comments relative to Appendix C.

- Regarding your Theorem 5.2, having a good ϵ-precision thus requires to have a number of samples b large enough (as $b = c_b T$) under the constraint to be smaller than n. Is it really achievable and not too restrictive in practice ?

- (Theorem 5.2/proof) See my comments relative to Appendix C.
- (Lemma 5.8) I understand the connection between your algorithm and those presented in [62]. However, I think a brief proof would be welcome, particularly to explain why you do not obtain the same coefficients in the right-hand side of (5.20) as in the original paper.
- (Lemma 5.10) Similar comment as for Lemma 5.7, especially the fact you need to justify the lower-boundedness of $\{\Psi_r(\mathcal{Q}^k)\}$ to be able to invoke Ψ_r^*.
- ((Lemma 5.10)/Theorem 5.3 proofs) See my comments relative to Appendix D.
- "From (5.25), to achieve".

6 Experimental Setup

- (page 17) including pointwise [...] methods" add references.
- (page 17) "regularization term $h(\boldsymbol{z}) = \lambda\||\boldsymbol{z}| - 1\|_1$ (the first z was not in bold).
- (page 17) "are provided in the Supplementary Materials E".
- (page 17) "to observe convergence behaviours".
- (page 18) Is there a particular reason why you are using a step size of 0.1τ rather than τ when updating Λ_k?
- It would have been interesting to also display, in Tables 6.2 and 6.3, the performances of DualHash-StoRM.
- I think you could try increasing the size of a few figures (especially 6.4).
- (page 22) "helps to produce high mAP".

7 Conclusion and Discussion

- "including image reconstruction".
- "and binary neural networks" (no comma).

8 Appendices A & B

- (page 28) See my first comment of preliminaries section about closeness vs lower-semicontinuity.
- (page 28) In the equality just before Prop A.2, one of the x is not in bold.
- (Prop A.2) In (iii), I would rather write $\hat{\partial} f_1(\overline{\boldsymbol{x}}_1), \hat{\partial} f_2(\overline{\boldsymbol{x}}_2)$ instead of $\hat{\partial}_{\boldsymbol{x}_1} f_1(\overline{\boldsymbol{x}}_1), \hat{\partial}_{\boldsymbol{x}_2} f_2(\overline{\boldsymbol{x}}_2)$
- (Prop A.3) I agree with the results of (i), (ii) and (iii) but the corresponding lemma/article (Lemma 5.2 [45]) doesn't not appear to be the correct reference.
- (Appendix B) I didn't really understand how you deduce, from the expression of L_s that L_F is n-independent. I would rather say that L_F can be bounded by a constant independent of n. Following one of my previous comment, I don't really like the use of $\mathcal{O}$ notation speaking of constants.

9 Appendix C

The computational part of the proofs is generally carried out correctly, with only a few minor errors. However some justifications are missing or confused particularly, the way you pass to the conditional expectation.

- (page 29) Notations $\mathbb{E}_{\xi^{[k]}}$ and $\mathbb{E}_k$ (rather written $\mathbb{E}_i$) introduced in the main paper seems to correspond to the same object. Also I would place this notation part in Section 5 instead (see my 3rd point of "Complexity analysis section).

- (Lemma C.1) I would replace $\nabla f(\boldsymbol{z})$ by a generic direction $\boldsymbol{d} \in \mathbb{R}^q$, this doesn't change the proof. The reason is that, in the proof of Lemma 5.5, you apply (C.1) by considering $\boldsymbol{G}^k$, which is not a gradient strictly speaking (but merely an oracle).

- (Lemma 5.5/proof) To better justify (C.3), simply say you apply Lemma C.1 to $F(., \boldsymbol{B}^k)$ considering $\boldsymbol{x} = \boldsymbol{x}_k, \boldsymbol{y} = \boldsymbol{y}_k, \eta = \eta_k, \boldsymbol{d} = \boldsymbol{G}^k$. The rest of the proof appears correct to me.

- (Lemma 5.6/proof) In (C.7) you don't pass to conditional expectation correctly ; this should be be $\mathbb{E}_{\xi^{[k+1]}}$ before every terms. Fortunately, this doesn't seem compromise your proof to the extent that the "wrong" terms in $\mathbb{E}_{\xi^{[k]}}$ in your inequality seems measurable (to check) and can therefore be written without a conditional expectation, for example $\cancel{\mathbb{E}_{\xi^{[k]}}[\Delta_{\boldsymbol{x}}^k]} \rightarrow \mathbb{E}_{\xi^{[k+1]}}[\Delta_{\boldsymbol{x}}^k] = \Delta_{\boldsymbol{x}}^k$. Moreover, remind the reader you also use $\mathbb{E}_{\xi^{[k+1]}}[\|e_k\|^2] \leq \sigma^2/b_k$ (to justify).

  Finally, to reach the inequality after (C.7), you need to invoke Lemma C.2. For this reason, I would place Lemma C.2 before Lemma C.1.

- (Lemma C.2/proof) Before the calculus relative to $K_5 - K_3 - C_3 - C_4$, remind the reader equalities $K_5 = \frac{1-\alpha_k}{\eta_k} - L_F, C_3 = K_4, \overline{L}_F = 2L_F + K_3 + K_4$. In the calculus of $C_4 - K_6$, the 3rd line is even an equality. Moreover, this should be β_k instead of β everywhere.

  Regarding the last part, delete the two "≥ 0" after C_B and $\frac{2}{L_F}C_B$ expression respectively. Otherwise, this would be a logical error as, at this stage of the proof, you have not yet shown that C_B is positive. Moreover, I found $\tau \in \left(0, 6(\frac{\sqrt{56}-2}{13})\delta\frac{1}{L_F}\right)$. Finally, could you provide additionnal details to justify why $C_B > 0$ ensures $C_2 > 0$?

- (Relation (C.8)) Remind the reader the expression of $\boldsymbol{d}^k$ (this can be throughout a reference to an equation). Regarding your definitions relative to StoM, do we have $\boldsymbol{G}^k = -\boldsymbol{d}^k$? If not, how do you define $\boldsymbol{d}^k$ otherwise (in section 2, you only remind the update's rule relative to StoRM) ? Furthermore, it is not clear how you define $\boldsymbol{d}_{\boldsymbol{x}}^k, \boldsymbol{d}_{\boldsymbol{B}}^k, \boldsymbol{d}_{\boldsymbol{\Lambda}}^k$.

- (Lemma C.3) In (C.10), I think the correct expression is $S_4^k = 6(L_F^2\beta_k^2 + 2\eta_k^{-1}\alpha_k^2)$. To maintain consistency in the structure of your terms, I would rewrite $S_1 = 4(\tau^{-2} + 3)$.

- (Lemma C.3/proof) "is deduced from the $\underline{L_F}$-smoothness of $\underline{F(., \boldsymbol{B}^k)}$".

  To justify (C.15), I would mention inequality $(a + b)^2 \leq 2a^2 + 2b^2$ just after the F smoothness (and not after the inequality).

  "~~yield~~ conduct to (C.9)."

  To justify (C.16), I would have mentioned Lemma 5.3 directly instead of (5.6).

- (Lemma 5.7/proof) Remind the reader η_k is constant to justify $S_3^k = S_3, S_4^k = S_4$. Since all the terms are positive, I think it would be more appropriate to bound (C.9) using $\max\{S_1, S_2, S_3, S_4\}$ instead of $\max\{S_1, S_2\} + \max\{S_3, S_4\}$.

When you sum over $k = 1, ..., T$, term $(3 + L_\Delta)\frac{\sigma^2}{b_k}$ is incorrect and should be $\frac{(3+L_\Delta)\sigma^2}{L_F T}\sum_{k=1}^{T}\frac{1}{b_k}$.

In (5.17), in the upper bound, you thus should obtain $\frac{C_\sigma \sigma^2}{T}\sum_{k=1}^{T}\frac{1}{b_k}$ with $C_\sigma = \frac{3+L_\Delta}{L_F}$.

- (Theorem 5.2/proof) The previous mistake propagates into the proof of Theorem 5.2.

10 Appendix D

The strengths and weaknesses of the proofs contained in this section are similar to those in the previous appendix.

- (Lemma 5.9/proof) I don't really catch the interest of intermediate variable s here. You should write (D.1) directly. Moreover, (D.1) is a simple direct consequence of Lemma 5.4 combined with Lemma 5.8. Why make things more complicated with intermediate equations (5.9), 5.7) ?

  In (D.2), I don't understand where the term $(C_4 - K_6)\Delta_x^k$ comes from.

- (Lemma D.1) In Inequality of Lemma (D.1) and after a quick check of the proof of Lemma 8 of [49], I think this should be $\mathbb{E}_{\xi^{[k+1]}}[\|\boldsymbol{e}^k\|^2]$ instead of $\mathbb{E}_{\xi^{[k]}}[\|\boldsymbol{e}^k\|^2]$ (similar confusion as in the proof of (C.7)) . If it is so, you can hardly consider $\mathbb{E}_{\xi^{[k]}}[\|\boldsymbol{e}^k\|^2]$ as metric for the rest of your analysis. To avoid, this kind of problem, I think you should work with $\mathbb{E}[\|\boldsymbol{e}^k\|^2]$ metric instead and present an "in-expectation" version of Lemma D.1 i.e. of the form of :

  $$\mathbb{E}[\|\boldsymbol{e}^{k+1}\|^2] \leq ...\mathbb{E}[\|\boldsymbol{e}^k\|^2] + ... + ...\mathbb{E}[\Delta_{\boldsymbol{x}}^{k+1}].$$

  The resulting proof should therefore not undergo any significant changes.

  That is why, in the remaining proofs/results to review, I will assume that the inequalities are passed to the expectation rather than conditional expectation.

- Why do you keep $|\mathscr{I}_{k+1}|$ instead of using notation b_{k+1}?

- (Lemma D.2/proof) For the second inequality of (D.7), the multiplicative factor in front of the first term should remain $\frac{1}{8(1-\rho_k)\eta_1 L_f^2}$ instead of $\frac{1}{8\eta_1 L_f^2}$, unless you are able to prove that $\mathbb{E}[\|\boldsymbol{e}^k\|^2] - \mathbb{E}[\|\boldsymbol{e}^{k+1}\|^2] + \frac{2\rho_k^2\sigma^2}{|\mathscr{I}_{k+1}|} \leq 0$. This propagates into the rest of the proof.

  Using (D.3) and more specifically $\rho_k \geq 8\eta_1\eta_k L_F^2$, I obtain $\frac{\rho_k}{8\eta_1 L_F^2} - \frac{\eta_k}{2} \geq \frac{\eta_k}{2}$.

  As in the proof of Lemma 5.7, you here need to justify that $\{\mathbb{E}[\Psi_r(\mathscr{Q}^k)]\}$ is non-increasing.

- (Lemma D.4) I would change notation C_δ^1 by C_δ (you did this in the proof).

- (Lemma D.4 proof) I would justify the first term in (D.12) in a similar way to what you did for the second one, this time by invoking the second inequality in (5.23) and the fact that $\tilde{S}_4 \geq S_1, S_2$. Moreover, I think this is $\tilde{S}_4$ everywhere instead of S_4.

  It appears more logical me to substitute (D.13) into (D.12) before passing to expectation.

  In (D.15), I think this is $\frac{2(C_\delta^1+4)}{\eta_1}$ instead of $\frac{2(C_\delta^1+4)}{S_4\eta_1}$.

  The end of the proof as well as those of Lemma 5.10, Theorem 5.3 are based on the potential new (D.4) inequality. To be reworked accordingly.

References

[1] H. H. Bauschke and P. L. Combettes. *Convex Analysis and Monotone Operator Theory in Hilbert Spaces*. CMS Books in Mathematics. Springer, Cham, 2 edition, 2017.

[2] D. P. Bertsekas. Nonlinear programming. *Journal of the Operational Research Society*, 48(3):334–334, 1997.

[3] A. Repetti and Y. Wiaux. Variable metric forward-backward algorithm for composite minimization problems. *SIAM Journal on Optimization*, 31(2):1215–1241, 2021.